Publication 1:

Title	Customer Churn Prediction: A Systematic Review of Recent Advances, Trends, and Challenges in Machine Learning and Deep Learning.
Harvard Reference	Imani. M, Joudaki. M, Beikmohammadi. A, Arabnia. H (2025) 'Customer Churn Prediction: A Systematic Review of Recent Advances, Trends, and Challenges in Machine Learning and Deep Learning.' *Machine learning & knowledge extraction*
Link	https://www.researchgate.net/publication/390255151_Customer_Churn_Prediction_A_Systematic_Review_of_Recent_Advances_Trends_and_Challenges_in_Machine_Learning_and_Deep_Learning
Modelling	Ensemble methods (XGBoost, LightGBM), Deep Learning (CNN, LSTM)
Preprocessing techniques	Feature scaling, categorical encoding, balancing imbalanced datasets.
Datasets used	Multiple telecom and banking datasets (review-based, including public datasets like IBM Telco)
Key takeaways	• Ensemble methods dominate churn prediction. • Deep learning increasing applied. • Preprocessing is critical for imbalanced churn data.

Publication 2:

Title	Research on Customer Churn and Model Interpretability Analysis
Harvard Reference	Peng. K, Peng. Y, Li. W (2023) 'Research on Customer Churn and Model Interpretability Analysis' *PLoS One.* 18(12).
Link	https://pmc.ncbi.nlm.nih.gov/articles/PMC10707658/
Modelling	Logistic Regression, Random Forest, Gradient Boosting, Neural Networks, interpretability analysis with SHAP values.
Preprocessing techniques	Data cleaning, feature encoding, sample methods (SMOTE, under sampling)
Datasets used	Kaggle bank churn dataset
Key takeaways	• Handling class imbalance is crucial. • Interpretability (SHAP) adds business value. • Ensemble methods outperform single classifiers.

Publication 3:

Title	An Analysis on Classification Models for Customer Churn Prediction
Harvard Reference	Mouli, K. C., Raghavendran, Ch. V., Bharadwaj, V. Y., Vybhavi, G. Y., Sravani, C., Vafaeva, K. M., ... Hussein, L. (2024). 'An Analysis on Classification Models for Customer Churn' prediction. *Cogent Engineering*, *11*(1).
Link	https://www.tandfonline.com/doi/full/10.1080/23311916.2024.2378877
Modelling	Logistic Regression, Decision Trees, Random Forest, Support Vector Machines (SVM), Naïve Bayes.
Preprocessing techniques	Feature selection, categorical encoding, normalization, and handling class imbalance.
Datasets used	Telecom churn datasets (proprietary and benchmark datasets)
Key takeaways	• Ensemble methods outperform single classifiers. • Preprocessing strongly influences accuracy. • Logistic Regression remains interpretable but less accurate compared to Random Forest/SVM.

Dataset Publication:

Name	Banking Customer Churn Prediction Dataset
Harvard Reference	Badole. S (2024) Bank Customer Churn Prediction Dataset. *Kaggle.*
Link	https://www.kaggle.com/datasets/saurabhbadole/bank-customer-churn-prediction-dataset/data
Number of instances	10 000 customer records
Number of feature columns	12-14 features including demographics, account info and financials
Number of classes	2 classes (Exited = 1, Retained = 0)
Target range	Exited (binary churn label)
Origin	Real-world European bank customer data, made publicly available via Kaggle